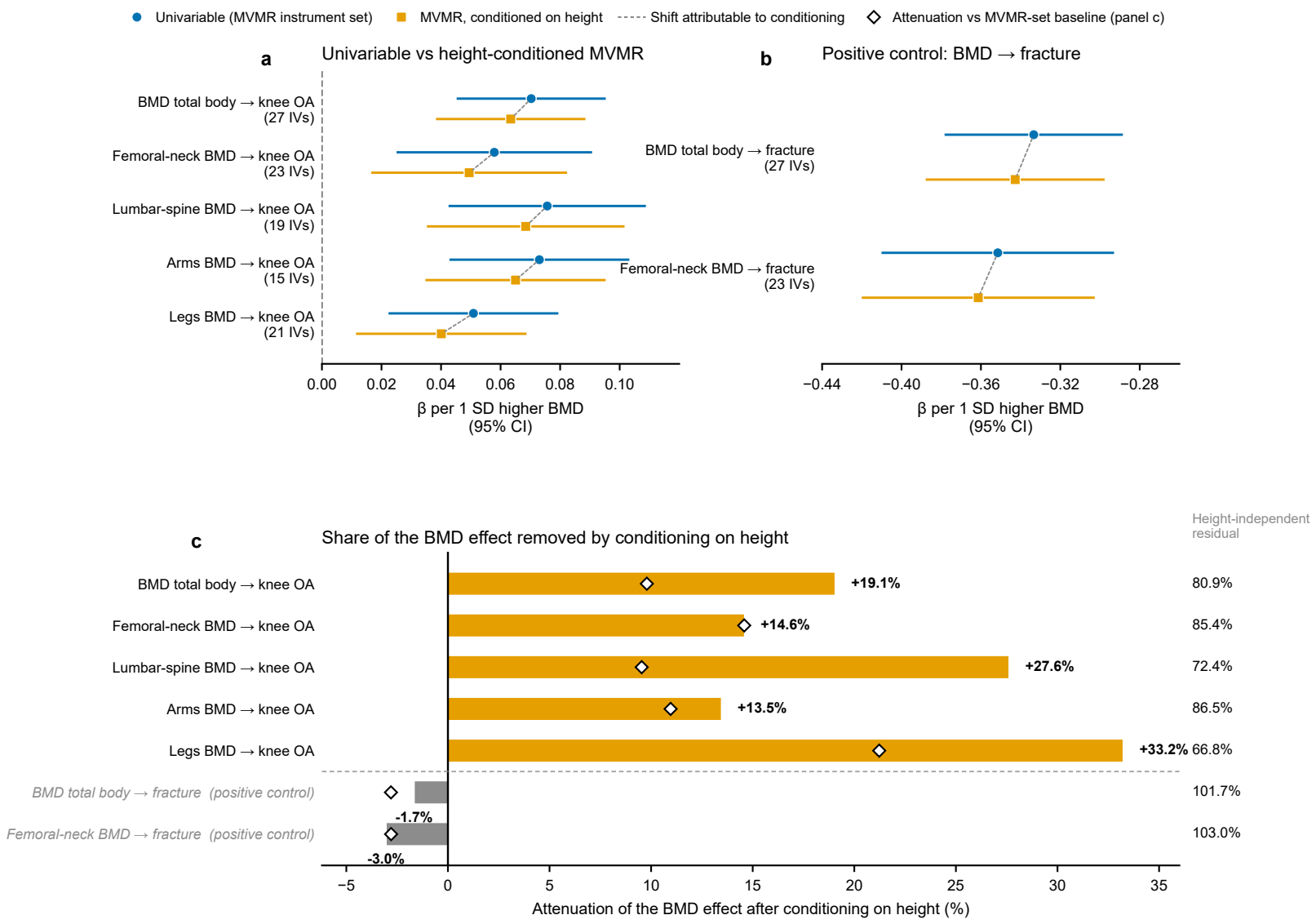

Univariable (MVMR instrument set)
MVMR, conditioned on height
Shift attributable to conditioning
Attenuation vs MVMR-set baseline (panel c)
a
Univariable vs height-conditioned MVMR
BMD total body → knee OA (27 IVs)
Femoral-neck BMD → knee OA (23 IVs)
Lumbar-spine BMD → knee OA (19 IVs)
Arms BMD → knee OA (15 IVs)
Legs BMD → knee OA (21 IVs)
0.00
0.02
0.04
0.06
0.08
0.10
β per 1 SD higher BMD (95% CI)
b
Positive control: BMD → fracture
BMD total body → fracture (27 IVs)
Femoral-neck BMD → fracture (23 IVs)
−0.44
−0.40
−0.36
−0.32
−0.28
β per 1 SD higher BMD (95% CI)
c
Share of the BMD effect removed by conditioning on height
Height-independent residual
BMD total body → knee OA
+19.1%
80.9%
Femoral-neck BMD → knee OA
+14.6%
85.4%
Lumbar-spine BMD → knee OA
+27.6%
72.4%
Arms BMD → knee OA
+13.5%
86.5%
Legs BMD → knee OA
+33.2%
66.8%
BMD total body → fracture (positive control)
-1.7%
101.7%
Femoral-neck BMD → fracture (positive control)
-3.0%
103.0%
−5
0
5
10
15
20
25
30
35
Attenuation of the BMD effect after conditioning on height (%)